

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1
24-10-2023

EDMS 01-300-1

Technical specification for
MV Primary Switchgear Remote Terminal Unit

Issue: Oct-2023 / Rev- 1

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1
24-10-2023

CONTENTS

<u>Clause</u>	<u>Description</u>	<u>Page</u>
1	SCOPE	7
2	REFERENCE	7
3	APPLICABLE STANDARDS	7
4	GENERAL REQUIREMENTS	8
5	RTU ARCHITECTURE	9
6	RTU Functions	10
7	RTU DESIGN	12
8	COMMUNICATION	17
9	CYBER SECURITY	19
10	RTU CABINET	21
11	RTU POWER SUPPLY	23
12	RTU Batteries	23
13	TESTING AND INSPECTION	24
14	TRAINING	25
15	AFTER SALE SERVICE	26
16	WARRANTEE	26
17	MARKING	26
18	SUBMITALS	26
19	OPTION LIST	27
20	GURANTEE TABLE	28

EEHC DISTRIBUTION MATERIALS SPECIFICATION MV Primary Switchgear RTU
--

EDMS 01-300-1 24-10-2023

GLOSSARY:

EEHC: Egyptian Electricity Holding Company.

EDMS: Egyptian Distribution Material Specifications.

RTU: Remote Terminal Unit.

DP: Distribution Point

CPU: Central Processing Unit

DI: Digital Input

DO: Digital Output

SP: Single Point

DP: Double Point

MV: Medium Voltage.

A.C.: Alternating Current.

D.C.: Direct Current.

IEC: International Electrotechnical Commission.

I/O: Input / Output

NO: Normal Open.

NC: Normal Close.

ADMS: Advanced Distribution Management System.

SCADA: Supervisory Control and Data Acquisition.

LAN: Local Area Network

WAN: Wide Area Network

RSTP: Rapid Spanning Tree Protocol

SOE: Sequence of Events

IP: Ingress Protection.

IK: Degree of Protection against Mechanical Impact.

C.B.: Circuit Breaker.

IP: Internet Protocol

...EDC: Electricity Distribution Company.

TCP: Transmission Control Protocol.

HMI: Human Machine Interface.

EEHC DISTRIBUTION MATERIALS SPECIFICATION MV Primary Switchgear RTU
--

EDMS 01-300-1 24-10-2023

RTC: Real time Clock.

GPS: Global Positioning System

SPST: Single Pole Single Throw contact.

RBAC: Role Base Access Control.

TLS: Transport Layer Security.

SSH: Secure Shell.

NTRA: National Telecom Regulatory Authority.

PRP: Parallel Redundancy Protocol.

HSR: High-availability Seamless Redundancy.

RSTP: Rapid Spanning Tree Protocol.

SFTP: Secure File Transfer Protocol.

HTTPS: Hypertext Transfer Protocol Secure.

LDAP: Lightweight Directory Access Protocol.

RADIUS: Remote Authentication Dial-In User Service.

DNP: Distributed Network Protocol.

SNMP: Simple Network Management Protocol.

SNTP: Simple Network Time Protocol

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

Definitions:

Alarm: An alarm is a signal indicating the system has experienced a change or an unusual situation.

Analogue signals: analogue values that fall within a range.

Digital signals: signals that can only exist in two states are known as digital signals.

Ethernet: Ethernet is the traditional technology for connecting devices in a wired local area network (LAN) or wide area network (WAN)

Event: is a signal for every incident that takes place within the system.

Field Equipment: equipment that the remote control system will control, such as circuit breakers, disconnectors,....etc.

Protocol: Rules and standards that deal with the exchange of information between computers and other devices. These have been defined to make the reliable, effective transfer of information possible.

Remote Terminal Unit: It is a device for acquisition of data and execution of commands by remote access.

Router: Is a device that forwards data packets along networks. A router is connected to at least two networks, commonly two LANs or WANs or a LAN and its ISP's network. Routers are located at gateways, the places where two or more networks connect.

SCADA System: A system based on electronic devices that makes it possible to remotely supervise and control any type of facility.

Inspection: Activities such as measuring, examining or testing any characteristic of a product or service, comparing obtained results with specified requirements, in order to determine if each characteristic complies with these.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

1. SCOPE

This specification specifies the minimum technical requirements for design, supply, engineering, manufacturing, testing, inspection, packing, and performance of the Remote Terminal Unit (RTU) intended to be installed in the primary distribution substation (DP) required achieving the required signal list. The RTU shall provide control, monitoring, and data acquisition function and interface to the Distribution management system (DMS).

2. REFERENCE

The RTU equipment's should comply with IEC listed hereafter. In no way does this specification supersede the relevant codes except when the requirements herein are more stringent. All cases of conflicts should be immediately brought to the tenderer's attention.

3. APPLICABLE STANDARDS

Unless otherwise specified, the RTU should comply with the latest edition of IEC publications:

3.1.	IEEE C37.1	IEEE standard for SCADA and Automation Systems
3.2.	IEC 61850	Communication networks and systems for power utility automation
3.3.	IEC 60870-5-101	Tele-control equipment and systems – Part 5-101: Transmission protocols – Companion standard for basic tele-control tasks
3.4.	IEC 60870-5-103	Tele-control equipment and systems - Part 5-103: Transmission protocols - Companion standard for the informative interface of protection equipment
3.5.	IEC 60870-5-104	Tele-control equipment and systems – Part 5-104: Transmission protocols – Network access for IEC 60870-5-101 using standard transport profiles
3.6.	IEEE 802.3	IEEE Standard for Ethernet
3.7.	IEC 61000-4	Electromagnetic compatibility measurements and testing.
3.8.	IEC 60255	Measuring relays and protection equipment
3.9.	IEC 60947-5-1	Control switches (low-voltage switching devices for control an auxiliary circuits, including contactor relays). - Part 1: General requirements
3.10.	IEC 60529	Degrees of protection provided by enclosures (IP Code).
3.11.	IEC 60086-2	Primary batteries - Part 2: Physical and electrical specifications

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

3.12.	IEEE 802.11	Information Technology
3.13.	IEC 62443-4-1	Security for Industrial automation and control systems – part 4-1: Secure product development lifecycle requirements

In case of any deviation from the listed standard, it should be indicated in the list of deviations list submitted by the tenderer.

4. General Requirements :

4.1. General

The remote terminal unit shall be able to communicate with the distribution control center (DCC) to exchange data.

The RTU shall collect circuit breakers status signals, alarms, and measurements from the switchgear and send them to the distribution control center (DCC), as well as issue commands to the circuit breakers in response to a command delivered from the DCC.

The data required by the DMS is collected from the switchgear, intelligent protection devices, smart meters, existing battery chargers, RTU battery chargers, DP doors; limit switches, sensors, auxiliary relays, auxiliary switches, and firefighting status.

4.2. ENVIRONMENTAL CONDITIONS

The electrical and mechanical properties of the required RTU and its peripheral devices should be guaranteed under the following environmental conditions:

Ambient temperature	-5°C to 50°C
Average relative humidity	95 % non-condensing
Altitude	Up to 1000 m above sea-level

4.3. GENERAL TECHNICAL REQUIREMENTS

- The RTU shall provide remote control, monitoring and data acquisition and interface to the SCADA system.
- Measured data shall be sent to DMS periodically, by demand, and when exceeding deadband values. Additionally, by Exception for status and alarm data shall be sent when status changes or alarm occurs.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

- The RTU shall be able to communicate with internal devices (I/O Modules, GPS, Communication modems, switches, etc.....) and external field devices (IEDs, Meters, switchgear, Battery charger, RTU battery charger, etc.....).
- The RTU shall acquire and process the data from the field devices and control them. It shall send the processed data to the control center and receive commands and requests from it.
- EDC reserves the right to require a documented demonstration of the RTU capabilities to support DMS SCADA functionality.
- The RTUs should have an architecture that supports convenient installation, maintenance and expansion features (modularity).
- The RTU should include all hardware, software and firmware necessary to meet the I/O point requirements including input and output cards and output relays.
- The RTUs configuration should include two central processing unit (CPU) modules (as option as perEDC requirements).
- The RTU shall have a Built-in webserver for commissioning, upload and download the configuration program and maintenance with local and remote access.
- RTU service life time shall not less than 20 years.
- The unit should be extendable & including 25% spare for any future expansion.

5. RTU Architecture

- The RTU system shall provide system redundancy and shall avoid any single point of failure except Rack and DI/DO modules.
- The redundancy shall be accomplished in the system via hardware, cabling, ports, software, etc.....).
- The RTU shall guarantee redundant supply module inside the RTU in which one power supply malfunctions, a second power supply takes over.
- Upper level communication with the DCC:
 - The communications between DCC and RTU cabinets are through IEC 60870-5-104/101 protocol to visualize and control the switchgear.
 - The RTU shall have two redundant CPUs with hot-standby configuration. The DCC will communicate with the RTU through one floating IP address. The link of the Ethernet port of the hot CPU shall be monitored. If the RTU hot don't detect traffic in his port, it shall switch to the standby CPU to avoid fails of communication when cables are unplugged accidentally or prevent fail of the network.
- Lower level communication with field devices:
 - The RTU shall use two Ethernet ports using Bonding protocol (so, the two ports act as single logical link with only one IP address) to receive the required data from the IEDs. The communication between the RTU cabinet and the IED's is through two redundant IEC 61850 Ethernet switches using (RSTP and/or HSR and/or PRP).

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

- The RTU shall retrieve analogue values (voltage, current, frequency, power factor, active power, reactive power, active energy, reactive energy) and status through digital inputs from the IEDs installed on each LV cabinet of MV cells of the switchgear. The IEDs shall be connected in loop to communicate with the RTU over TCP/IP using IEC 61850 protocols.
- In case of the IEDs are not supports IEC 61850 communication protocol the RTU shall communicate with the existing IEDs with the available communication protocols.
- The RTU communicate with the Battery Charger uses protocol Modbus TCP/IP through the two switches.
- The RTU communicate with the switches, router, NTP server through SNMP (V3) protocol or equivalent to collect and organize information about managed devices.
- Hardwired I/Os connection to RTU (complementary status, IED's watchdog, MCB position, site signals such as, door open, firefighting operation ...etc).

6. RTU Functions

- The RTU shall have the basic functions including remote control, measurement, detecting event/alarm, recording, information points from devices, relays and/or IED's data communication, remote/local switching, internal clock, time synchronization, time stamp/tag, data holding, diagnosis of itself in failures and etc.
- RTU shall be multi- center multi-protocol.
- Receiving and processing digital and analog control commands from the master station or SCADA.
- RTU shall support Sequence of Events (SOE) feature with 1 msec. resolution with time stamping with capacity shall be at least 5000 events to be accessed from the ADMS and at site.
- During the communication failure between the DCC and the RTU, the RTU shall store at least 5000 events and 5000 analogue values and to be accessed from the ADMS and at site.
- RTU shall have logic functions to allow customized automation schemes and should be ready to activate an IEC61131-3 logic engine or similar.
- Interface shall be preferably web-based.
- The RTU shall have the capability to act as a NTP time server
- It shall be possible to implement the alarming.
- The RTU data/time shall not affected by the loss of the power supply.
- RTU shall act as a data concentrator for acquiring data from slave RTU's and exercising supervisory control on slave RTU's.
- RTU shall accept polling messages from at least two master stations simultaneously using separate logical databases for each master station and the control can be configured from one master station only.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

- RTU shall communicate simultaneously on all communication ports.
- The RTU shall have the option of redundant communication through the Ethernet ports.
- It shall be possible to export database (signal list) to excel or similar software application.
- The data transmission rates shall be not less than 9600 baud for serial ports and 10/100 Mbps for TCP/IP.
- RTU shall have the capability of automatic re-start after a power outage.
- RTU shall be scalable.
- The RTU shall be designed to communicate with a minimum 40 Electronic devices or IED's connected with 500 variables per device.
- The RTU shall be capable to calculate all logical functions needed by an Advanced Distribution Management System such as Power factor, Active power, Reactive power, Active and reactive energy. (as option as perEDC requirements)
- The RTU capable to interface with the programming laptop (Programming, troubleshooting, etc.).
- The software should be providing automatic restart of the RTU upon power restoration, memory parity errors, hardware failures, and manual request.
- The software should initialize the RTU and being execution of the RTU function without intervention by the SCADA/DMS master station.
- All requests should be reported to the SCADA/DMS.
- In order to easily support the system under continuously changing site conditions all protocol, configuration, firmware, and application data must be contained in easily programmable non-volatile memory such as flash EPROM.
- Updating the firmware and the configuration of the RTU shall be available from ADMS and by using maintenance tool on the site.
- Updating the RTU firmware shall not change the logic and data structure.
- The RTU design should be independent of any communication protocol that would impose restrictions on the flexibility of functionality of the RTU.
- Protocol changes should be accomplished by software/firmware changes only.
- It shall be possible to retrieve the configuration of an existing RTU without the need of the original configuration backup.
- Any control command from the DCC shall follow select check before operate (SCBO) procedure for control operation.
- The RTU shall have the following features :
 - Event time tagging to 1 m sec resolution and will have time synchronization capability with GPS
 - The RTU shall have self-diagnostic function for ascertaining soundness of the processors, memories, internal battery, I/O modules, power supply, communication to DMS, time synchronization, and CPU Redundancy.
 - Perform automatic initialization when power is restored after a power loss

EEHC DISTRIBUTION MATERIALS SPECIFICATION

MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

- In case of communication failure with the host equipment, the RTU shall have capability to store data (events / alarms) and shall continue to operate on a standalone mode independent of host.
- The RTU shall provide battery back-up to volatile memory in order to retain data during an extended loss of power.
- It shall be possible, from the engineering workstation, to simulate and monitor program execution or force a change in a definable variable at any RTU.
- The remote/Local switch:
 - The switch shall be changeover switches and covered for prevention of unintentional changeover and can be operated from the site only.
 - Each switch shall have double-status. Signal of both statuses shall be sent to the ADMS.
 - The switch defines behavior of the RTU as follows;
 - In Remote position: the DMS can control circuit breakers remotely. Open/Close command is accepted by the RTU, and RTU outputs a control signal to the target IED.
 - In Local position: Open/Close circuit breakers can be done in manual at site. The control command from DMS is not accepted by the RTU.

7. RTU Design

- The RTU shall accomplish a high-speed internal bus for the communication between the acquisition modules and the CPU.
- Double power-supply bus (redundant).

The RTU cabinet shall comprise from and not limited to the following component:

- Two central processing unit (CPU) (as option as perEDC requirements).
- Digital input Modules*
- Digital output modules*
- Radio Station
- 3G/4G Router. (as option as perEDC requirements.)
- Redundancy modules for the two DC supply sources
- Two switches (8 or 16 Ports). as perEDC requirements.
- NTP server - Global positioning system (GPS) (as option as perEDC requirements.)
- AC & DC MCBs for circuit's protection.
- Interface terminal blocks with the internal and external devices.

7.1. Central Processing Unit Module (CPU)

- Central processing unit (CPU) should handle all protocol emulation i.e. IEC60870-5-104/101, IEC 61850, MODBUS master ...etc., perform data acquisition, and control requests.
- The switching mode between the Hot and standby CPUs shall be HOT DATA mode which allow database of the standby CPU to be updated from the online CPU.
- The failover between the two redundant units doesn't require any manual intervention and

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

doesn't cause any interruption in the functioning of RTU.

- The CPU should accept commands from the master station, perform address recognition, assemble response message in accordance with the received command message, and transmit this message to the SCADA / DMS master station.
- The CPU should have RTC and provide interfaces for time synchronization from GPS and from DCC over IEC60870-5-104/101.
- The CPU should have user configurable routines/procedures to carry out connection establishment, link failure detections and reconnection after failures for dialup connectivity.
- The CPU should manage communication between all other function modules of RTU and should determine the integrity of the RTU.
- The processor should provide diagnostic information in the message structure that the SCADA / DMS should monitor.
- A flag should be set if the RTU perform a restart for any reason including power failure.
- The CPU should be programmable in a high language.
- It should be able to program the RTU and manage the RTU database from the RTU test set and download parameters and configuration data from the SCADA / DMS system.
- The RTU shall have SNMP protocol or equivalent for asset management (Router, GPS, Switches,).
- The status of the online, fail, synch, and internal battery status shall be displayed.
- RTU Communication:
 - Each CPU shall support different protocols to communicate with the SCADA/ADMS as a server/slave:
 - IEC 60870-5-104/101 as mandatory
 - IEC 61850
 - IEC 60870-5-101
 - DNP 3
 - Each CPU shall support different protocols to communicate with the RTU automation devices and protection relay as a client/Master:
 - IEC 61850 and Modbus as mandatory
 - IEC 60870-5-101/103/104
 - DNP 3.
 - Each CPU shall support different interfaces of communications:
 - Four Ethernet ports (Two for Remote connection and two for local connection)
 - At least two serial ports RS 485 for local connection.

7.2. I/O MODULE

- Each I/O module should be capable of interfacing with digital inputs, control output points and combination of point's types.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

- I/O modules should be replaced without reprogramming, redefinition of configuration parameters or rewiring.
- The RTU should include facilities for handling status input and control output points. Requirements for each type of I/O point.
- The required number of I/O points should be the responsibility of the contractor, considering 20 % spare of I/O points.
- It should be possible to expand the RTU capacity by an additional 20% of the initially delivered (including spares) I/O points by providing spaces for adding cards and terminations at future date.
- The status of the Run, fail, polarization voltage status shall be displayed.
- STATUS INPUTS
 - The input circuit of the status input modules should be optically isolated from external signal (input isolation), current limiting and bounce filtering.
 - The de-bounce period for each status input should be individually configurable to ensure digital filtering.
 - The input circuit shall have protection device for electrical surge, noise and etc. And also, it shall be considered to prevent an incorrect input signal caused by electrical surge, noise, and etc.
 - The module shall accept contact types which are SPST-NO/NC and SPDT (Single-Pole-Double-Throw).
 - Each input circuit should include a LED indicator next to the circuit terminations or displayed through LCD screen to show the status of the associated input contact.
 - The state of each status point should be reported to the SCADA / ADMS on a contention basis. That is, a status point should be not being reported unless the point state has changed from the last scan.
 - The RTU should also report the state of selected status point upon receipt of a demand scan request from the SCADA / ADMS.
- CONTROL OUTPUTS
 - The RTU should include on / off device control points to support control actions initiated from SCADA / DMS.
 - The RTU should perform on / off control actions using complimentary pairs of contact outputs.
 - The RTU should include momentary control outputs as required by the device being controlled.
 - Each output circuit should include a LED indicator next to the circuit terminations or displayed through LCD screen to show the status of the associated output contact.
 - The Digital outputs shall be single pole single throw (SPST) normally open (NO) contacts.
 - The controlled outputs should be capable of driving a load of (5) amps at the primary control voltage with provision for an additional NO contact. Besides, DI status of command execute acknowledgement.
 - External auxiliary control relays are not preferred but may be applied if integral

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

relays do not satisfy the above ratings, where these relays should be supplied by the contractor.

- All control point should follow select check before operate (SCBO) procedure for control operation.
- Point selection should be cancelled automatically following the completion of the control action, and reselection of the point should be required for subsequent control actions.

7.3. RTU Time and Date

- The RTU should have an internal real time clock (RTC) for data collection coordination and time tagging.
- The RTU synchronization should have at least two synchronization sources:
 - In case of using separate SNTP server: primary through GPS through SNTP protocol and secondary through the communication protocol with DCC.
 - In case of not using separate SNTP server: primary through the communication protocol with DCC and secondary from the RTU itself.
- The RTU RTC should be maintained within hundred (± 1) millisecond of the same time reference used by the respective SCADA/DMS.

7.4. Diagnostic Software

- Software should provide to continuously monitor operation of the RTU and report RTU hardware errors to the SCADA/DMS.
- The software should check for memory, processor and input/output errors and failures.

7.5. Quality Assurance

- The remote terminal unit (RTU) package should be designed, constructed and tested in accordance with NEMA, NEC and IEC standards.
- Prior to shipment, the remote terminal unit (RTU) package should be fully functionally tested at the factory, as an assembled unit with communications devices installed and operational to insure proper system operation.
- All pilot devices and controls should be tested to verify proper operation. Testing documentation should be furnished at the engineer's request.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

7.6. RTU Testing

- Type Test: it should comply the following standards:
 - IEC 60255- 4, 5 and 6 and IEC 60255-22.1, 2 and 3. & IEC 61000 - 4 -2 and 3.
- Routine Test: the RTU should pass the manufacture's standard routine tests in accordance with referenced standard.

7.7. Signal List

- Signal List of the RTU is attached.
- Signal List of the Switchgear as per EDMS 01-101-1 specification
- The Mapping of the signals list will be submitted after approval.

Signal List for DP RTU

Item	Signal Description	Signal Acquisition	Type of Data Point
Status			
RTU Cabinet	Main Comm. Link Failure		
	Backup Comm. Link Failure		
	RTU Failure		
	Redundant RTU Failure		
	Main Time Synchronization Failure		
	Backup Time Synchronization Failure		
	CPU 1 Online		
	CPU 2 Online		
	RTU Cabinet DC Supply 1 Loss		
	RTU Cabinet DC Supply 2 Loss		
	RTU Cabinet AC Aux Supply Loss		
	Fail of Power Supply of Rack		
	DCC Online		
	Switch ports inactive		
	RTU Cabinet Local - Remote Indication	Hard wired	DP
	RTU Cabinet Door Opened		
	Any MCB in RTU Fail		
RTU Battery Charger	RTU Battery Charger Float Mode	Communication	
	RTU Battery Charger HI-Rate Mode	Communication	
	RTU Battery Charger AC Loss	Communication	
	RTU Battery Charger Any MCB Fail	Communication	
	RTU Battery Charger AC over/under Voltage	Communication	

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

	RTU Battery Charger Failure	Hard wired	
	RTU Battery Charger DC Low Voltage	Communication	
	RTU Battery Charger Earth Fault	Communication	
Switchgear Battery Charger	Switchgear Battery Charger Float Mode	Communication	
	Switchgear Battery Charger HI-Rate Mode	Communication	
	Switchgear Battery Charger Any MCB Fail	Communication	
	Switchgear Battery Charger Failure	Hard wired	SP
	Switchgear Battery Charger AC Loss	Communication	
	Switchgear Battery Charger Earth Fault	Communication	
	Switchgear Battery Charger DC Low Voltage	Communication	
DP	DP Entrance Door Open	Hard wired	SP
	Fire Fighting Operate	Hard wired	SP
Commands			
RTU	RTU CPU Reset		
Measurements			
RTU Battery Charger	RTU Battery Charger AC Voltage	Communication	
	RTU Battery Charger DC Voltage	Communication	
	RTU Battery Charger DC Current	Communication	
Switchgear Battery Charger	Switchgear Battery Charger AC Voltage	Communication	
	Switchgear Battery Charger DC Voltage	Communication	
	Switchgear Battery Charger DC Current	Communication	

8. Communication

8.1. General Requirements

The connectivity between the distribution point and the DCC shall be achieved by:

- Private Point to Multipoint Radio System (VHF or UHF), as primary path
- Using 3G/LTE Data Service with one Telecom Operator, as a backup path. The Telecom Operator vendor shall be selected depending on the signal strength. The Cellular Router will be equipped with one SIM card for the selected Operator. Note: SIM card isEDC responsibility.

8.2. Telecom devices

- Industrial routers
 - The 3G/4G Router shall support LTE with bands (B1, B2, B8, B20, and B28).
 - Router should be capable of operating temperature up to 70 C°.
 - Router shall support by one SIM card at least.
 - The router shall support Ethernet and Fiber optic ports

EEHC DISTRIBUTION MATERIALS SPECIFICATION

MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

- The router with SIM card shall have its antenna. The length of antenna depends on the site.
- Nominal voltage 24 V-DC.
- RF Modem
 - RF modem shall connect with the mast which installed on the rooftop.
 - VHF/UHF frequency.
 - Channel spacing 25 kHz and also configurable for operation in (12.5 kHz, and G.25 kHz).
 - Modulation: QAM and any other modulation techniques.
 - The transmitted output power at least: 6 w to 30w.
 - Direction antenna (yagi) with frequency range (130-150) MHZ.
 - The mast length depends on the location of the site and the distance to the control center.
 - The RF modem shall support Ethernet port.
 - Should be capable of operating temperature up to 70 C°.
 - The RF modem shall be Simplex, half duplex.
 - The RF modem shall be protected from high temperature and high VSWR.
 - The speed: from 60 up to 120 kbps.
 - The firmware shall be upgraded locally and remotely over the air.
 - The coaxial cable with all wanted accessories and its length depend on the situation on the site.
 - Nominal voltage 14-24 V-DC

8.3. Communication with SCADA

- The RTU shall have at least 2 communication ports which can be available to work in parallel. Each port shall be able to support individually the communication protocol of IEC60870-5-101/104, IEC61850, Modbus and others. The communication medium and protocol shall be set from the DMS or set by using local maintenance tool. For maintenance port, RS232, USB, or equivalent connection, as a minimum, shall be provided for interface between the RTU and local maintenance tool.
- The RTU shall be able to communicate with the SCADA on 2 channels.
- The RTU shall accept communication with at least 2 SCADA simultaneously.

8.4. Protocol

- The RTU shall comply with IEC 870-5-104/101, IEC61850 and Modbus as mandatory in addition IEC 870-5-101, IEC61850, DNP3.0 protocols if available. The RTU shall support Secure Authentication according to IEC 62351-5/3, and provide the data base as an IEC 61850 data model.
- Communication protocols of SCADA shall be supported by authentication and encryption.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

- The RTU shall comply with both server mode for the SCADA and client mode for devices in the substations.

8.5. Transmission

- The communication system with the control center shall be based on Radio system and 3G/4G or according to telecom study.

8.6. Transmitted data

- The RTU shall transmit to the SCADA all the status and measurement. Each data shall be individually configurable to be sent or not to the SCADA.
- The measurement shall be spontaneously sent to SCADA according to configuration of:
 - Threshold
 - Dead band

9. Cyber Security

- In order to secure all controls and data acquisition, the RTU shall be designed to be compliant with NERC, IEC 62443-4-1 and IEC62351 requirements. The RTU shall support secure access based on RBAC.
- Authentication by a Radius or LDAP server shall be provided.
- Local and remote access connection shall be secured for maintenance (locally and remotely) with HTTPS or SFTP or SSH protocols.
- The Firmware of the RTUs shall be digitally signed through embedded crypto-chip on the CPU which provides secure for uploading the firmware to the RTU.

9.1. Future proof design

- The RTU shall support remote firmware updates
- The RTU shall be evolutive in order to be compatible with a full centralized RBAC management in compliance with IEC 62351-8

9.2. Hardening

- Device hardening
 - Disabled or unused functionality shall not compromise security.
 - Unnecessary services and programs shall be removed. If removal is not possible, the unnecessary services and programs shall be disabled.
 - The RTU must provide mechanisms to prevent illegitimate firmware to be loaded into the system
- Interface minimization
 - Each interface shall support only the data types and protocols needed to meet the functional requirements.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 01-300-1
MV Primary Switchgear RTU	24-10-2023

- Unused interfaces and ports shall be removed. If removal is not possible, the unused interfaces and ports shall be disabled.
- A complete list of supported data types and supported communication protocols per interface shall be provided.
- All hardware interfaces that are used for programming or debugging shall be completely removed after production.
- Account hardening
 - The RTU shall not contain active default, guest and anonymous accounts.
 - All remote access to root accounts on the RTU shall be disabled.
 - All Vendor-owned accounts where feasible shall be removed.
 - The list of all accounts on the RTU shall be provided.

9.3. Compliance with security standards

- The RTU shall follow the IEC 62351 standards and at least: IEC 62351-5 : 2013 or IEC 62351-3

9.4. Configuration

- Access to the RTU by configuration tool shall be possible only through secured connection: HTTPS for Webserver or SSH for console and configuration tool.

9.5. Role Base access control (RBAC)

- The RTU shall support the implementation of Role-based Access Control in compliance with IEC 62351-8.
- The RTU should allow managing user authentication through a Radius server or LDAP (secured via TLS) according to IEC62351-8.
- It must be possible to carry out changes by configuration files through a secure way.
- It shall be possible to assign each role individual security credentials.
- A specific tool shall permit to configure the security policy, role and password.
- It shall be possible to bind roles to individual user accounts on the RTU.
- The minimum following function and data shall be controlled through RBAC:
 - Configuration files
 - Software update
 - User management
 - Executing program or shell command
 - I/O on local maintenance access

9.6. Management of Security passwords

- The RTU service application shall support individual user passwords.
- Passwords shall be stored together with a salt using an allowed cryptographic hash function.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

- The RTU service application shall enforce a high complexity of passwords.
- The RTU shall lock the access after several password failures.
- User Authentication
- The RTU shall authenticate the communication parties on the WAN interface using a challenge-response protocol based on message authentication codes The RTU shall terminate the connection if the user authentication fails.
- The RTU shall authenticate the communication parties on the Local Maintenance interface.
- It shall be possible to configure the RTU so that it blocks authentication requests, either temporarily or permanently, from an account after a number of failed login attempts. The number of failed login attempts and the time the account is blocked shall be configurable.

9.7. Central management of user account

- The RTU should allow managing user authentication through a Radius or LDAP server.

9.8. Security Log

- The RTU shall provide a local audit trail for all security events that occur.
- Log files shall be produced in Syslog format.
- Security events shall be logged locally in a dedicated security log or/and on a SYSLOG server.

9.9. Cyber Security Certificate

- The bidder must follow the NTRA & EG-Cert. cyber security approval requirements and the certification for the RTU system.

9.10. Documentation

- All released versions (hardware, firmware, software) of a device or product shall be uniquely identifiable.
- Exchangeable hardware modules shall be versioned separately.
- For Asset Monitoring the RTU shall provide firmware version, Serial number, and date of installation, via SNMPv3 Agent or equivalent.”
- The Protocol Implementation Conformance Statement as in IEC 62351 and IEC 60870-5-7 shall be provided on request.

10. RTU cabinet**10.1.1. General:**

- The RTU Panel shall be standalone panel.
- The RTU panel shall be mechanically divided into logical sections for RTU modules, switches, termination I/O, electrical power distribution,etc.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

- The measures of the RTU panel shall be not less than (700mm length - 700mm width - 2000mm highest).
- The panel shall be accessible from the front.
- It shall comply with IEC 60529, with an index of protection IP 54.
- RTU cabinet door shall equipped with safety latches door stop to hold the door in the open position
- The enclosure shall provide housing for the RTU hardware, power supply converters, communication modem, MCBs, auxiliary relays, terminal blocks and accessories.
- A 220 Vac outlet shall be provided for maintenance purpose.
- The enclosure should be Pad lockable and fabricated using 2 mm thick.
- The low voltage panel door shall be provided with key lock (master key).
- The panel shall be free standing, self-supporting structure and suitable for floor mounting.
- The panel shall support with lifting eyes for lifting which installed in the top.

10.1.2. Design:

- Internal layout design shall provide easy access of each component in the cabinet
- All wiring in the RTU shall be labeled directional from / to.
- The box shall be equipped with door limit switch.
- The panel shall be provided with a separate Local/Remote switch.
- Forced air cooling facility with thermostat shall be provided with IP filter according to the heat dissipation calculation which shall be submitted by the vendor.
- The cabinet shall have interior light which automatically switch on when the door open.
- It shall have thermostatically controlled heater to prevent condensation, if necessary.
- Each control/protection circuit shall be protected by appropriate MCB.
- The cabinet shall have interface terminal blocks between the switchgear and the RTU I/O modules. The terminal blocks shall accept appropriate CSA wires (knife type). The terminal blocks for output shall have fuses for protection.
- The Communication devices shall be installed on a separate rack with a separate forced air cooling fan.
- The RTU shall design to suitable with top cable entry.
- The cabinet shall have different entrances for power, interface, communication, and grounding cables.
- Reliable Earthing terminals with clamping screws or bolts shall be provided for connection of Earthing conductors. The connecting point shall be marked with the "protective earth" symbol, as indicated by the symbol 5019 of IEC 60417.

EEHC DISTRIBUTION MATERIALS SPECIFICATION

MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

11.RTU Power Supply

- The RTU cabinet shall have dual power supply sources:
 - 24 V-DC from the RTU battery charger (as main)
 - 24/110/220 V-DC from the SWG battery charger (as backup).
- The RTU shall provide over/under voltage protection for the two DC sources and the AC source.
- These two power supplies shall supply the RTU, communication modems, NTP server, auxiliary circuit, interface terminal blocks and accessories.
- The internal circuits for the power shall achieve the redundancy.
- The technical specification of the RTU Battery charger shall be attached.

12.RTU Battery

The batteries shall be vented nickel cadmium batteries according to EDMS 27-301 specification.

BATTERY CHARACTERISITICS:

- No of cells/battery 20.
- Nominal voltage 1.2 v/cell.
- Max. Voltage/cell 1.4 v.
- Nominal voltage 24 V/set.
- Max. Voltage 28 V/set.
- Rated capacity not less than $C_5 = 55 \text{ Ah}$, supply variable load, its type is (KPL) vented nickel cadmium (K), plastic container (P) and low rate of discharge (L) and shall provide a backup time not less than 8 Hours.
- The battery bank shall be able to provide upon discharge a current of 0.2C A for 5 hours to an end voltage of 1 V/cell.
- The cell can tolerate complete discharge with almost no permanent deterioration of capacity or life.
- Max short circuit current $10 \times C_5$ Amps.
- Charging characteristics: The battery can be charged by all following methods: -
 1. Constant current charging:
 - a) Standard charge 0.2 C_5 for 7-8 hours.
 - b) Fast recharge 0.3 C_5 for 2.5 hours followed by: 0.2 C_5 for 2.5 hours.
 - c) Trickle charge at about 1-2 m. A/A.H. capacity.
 2. Constant voltage:
 - a) Float voltage range at about 1.38-1.47 V/cell.
 - b) Boost voltage range at about 1.47-1.73 V/cell.
- The batteries shall be complete with the mounting racks.

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023
13. Testing and inspection:**13.1. Type Test:**

- All equipment shall be successfully type tested at EEHC approved laboratory in accordance to the latest relevant standard for the following:

13.2. Routine Test:

Routine tests of the RTU should be carried out according to latest relevant IEC standards by the manufacturer; a representative of Distribution Company will attend these tests before acceptance.

The.....EDC have the right to add any test during delivery to check the quality of the product.

...EDC reserves the right to visit the factory during manufacture of any or all items covered by this specification, for inspection of material or witness of tests.

Type test, FAT and Pre-commissioning test as SAT of the RTU shall be carried out separately from integration test of the DMS and trial operation. The Following table shows test items that shall be tested at least in each test.

Test	Description of Test	Type test	FAT	SAT
1	Check the operation of all Status input & Control output points of RTU	√	√	√
2	Check operation of all communication ports of RTU	√	√	√
3	Check for communication with DMS or DMS simulator	√	√	√
4	Test for downloading of RTU database from DMS or DMS simulator	√	√	√
5	Test for RTU time synchronization from DMS or DMS simulator	√	√	√
6	Test Power Supply Voltage Margin, Ripple Levels and Short Circuit Protection	√	√	
7	Test for RTU operation with DC power supply voltage variation	√	√	
8	Check for auto restoration of RTU on DC power recovery after its failure	√	√	√
9	Test for RTU diagnostic feature	√	√	
10	Test for RTU SOE (Sequence of Events Recording) feature	√	√	√
11	Test for RTU internal clock stability	√	√	
12	Test for Control Security and Safety for Control outputs	√	√	
13	Other functional tests as per technical specification requirements	√	√	
14	Point to Point test (between RTU & DMS) for all I/O points			√
15	EMI/EMS Test	√		
16	Dielectric Test	√		
17	Insulation resistance test	√		√

The following table includes the detailed routine test for the RTU cabinet and functions:

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

Panel's Type:		Panel's Name
DESCRIPTION OF TESTS: Pass(P) , Fail (F) , Un-completed (U) , Not available (NA)		
A	Electrical and Mechanical characteristics of RTU	Evaluation
1	Visual inspection tests: - All integrated component is compliant with the drawing - No physical damage - Check the nameplates and general labeling form/to. - Inspect cable entry glands, earth bar, cable support, cable clamp - Front door with lock. - Check CPU, DI/DO modules, Ethernet switches hardware, port counts, available spare ports, NTP server etc. - Verify electrical power supplies and ventilation. - Wiring installation, color coding, tagging. - Miniature Circuit Breakers, fuses, grounding, lights and outlets.	
2	RTU Functionality Tests - RTU Redundancy Tests. - RTU Digital Input tests. - RTU Digital Output tests. - RTU Modes of Operation (Local/Manual). - RTU Power Redundancy Test. - RTU Synchronization Tests. - RTU Communication Protocols - Cyber Security functions. - RSTP, HSR, PRP Tests	

14.Training:

....EDC reserves its right to claim a priceless training of its representatives on the vendor's factory, site, and lab.

The training should include and not limited to the following:

1. Installation.
2. Testing & Commissioning.
3. Troubleshooting.
4. Operating and Maintenance

EEHC DISTRIBUTION MATERIALS SPECIFICATION MV Primary Switchgear RTU	EDMS 01-300-1 24-10-2023
--	---

15. After sale Service:

The tenderer should have a licensed service center in Egypt covers all its deliverables.

....EDC reserves its right to claim a free after sale service all over the warranty period guaranteed by the tenderer that covers all its deliverables, after sale services should include and not limited to the following:

1. Technical support.
 2. Maintenance.
 3. Original Spare parts.
 4. Submittals.
- All the services, spare parts, materials and accessories will be needed during the after-sale maintenance should be guaranteed by the tenderer immediately and wherever required (Warehouse – Site – Lab.etc.)

16. Warrantee

- The supplier guarantees the RTU against all defects arising out of faulty design or workmanship, or of defective material for period of (2) years from date of delivery.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

17. Marking

17.1. The following data should be engraved or laser printed on an aluminum/stainless steel nameplate riveted on the Remote Terminal Unit Cabinet and located in a visible and legible place when the panel is installed showing the following information:

- Manufacturer's name
- Type/Model
- Serial Number and year of Manufacturing
- Distribution company name
- order number
- IEC standards

18. SUBMITTALS

18.1. Vendor shall provide the following with the quotation:

- Clause by clause compliance with the specification.
- All necessary drawings including the scheme drawing of the RTU, dimensions,

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1

24-10-2023

single line diagram of (control circuit, wiring diagram, voltage supply circuit, Alarms, communication.....etc.)

- Technical catalogues containing the technical data for all offered components of the unit should be submitted.
- Installation and maintenance instructions.
- Copy of type test report.
- A valid approved letter must be submitted from the Egyptian Electricity Holding Company
- A comprehensive list of manufacturer's recommended spare parts. The quantities offered should be adequate for the initial 2 years of operation. Firm price and delivery period shall be quoted for each item andEDC has the right to request the spare part list or not

18.2. Vendor shall provide the following after signing of purchase order:

- Details of manufacturing and testing schedules

18.3. Vendor shall provide the following with delivering:

- Operation (user) manual.
- Master Keys, Local/Remote key..... for operation.

19.OPTION LIST:

The following table should be filled/attached by ...EDC during tendering.

No.	Option	Needed	No Need
1	3G/4G Router. (default: Needed)		
2	Separate NTP server (GPS). (default: No Need)		
3	Capability of calculating power, Energy,....etc. (default: No Need)		
5	CPU Redundancy (default: No Need)		
4	No. of Cells per Switch gear should be defined according toEDC requirements: No. Of Cells []		

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023
20. TECHNICAL DATA SCHEDULE

The guarantee schedules should be submitted and the tenderer should fill in the attached guarantee tables.EDC reserves the right to reject any tender not accompanied with clear and complete guarantee tables.

GUARANTEE TABLES

No	Item	Vendor Spcs.
Environmental conditions		
1	Ambient temperature	°C
2	Average relative humidity	°C
3	Maximum altitude	m
RTU Functionality		
	The RTU is designed to accurately monitor and measure various parameters such as voltage, current, temperature, and pressure to provide real-time data on the condition of the equipment and systems it controls.	Yes/no
	It communicates seamlessly with the DCC, ensuring timely and reliable transmission of critical data, alarms, and status updates, thereby facilitating efficient decision-making and control.	Yes/no
	It supports secure remote access and configuration, enabling authorized personnel to make necessary adjustments, troubleshoot issues, and update settings without the need for physical presence at the site.	Yes/no
	The RTU is equipped with advanced diagnostics and self-testing capabilities to detect faults or anomalies within the unit itself, allowing for proactive maintenance and minimizing downtime.	Yes/no
	RTU support Sequence of Events (SOE) feature with 1 msec resolution with time stamping	Yes/no
	The RTU have the capability to act as a NTP time server	Yes/no
	The RTU data/time does not affected by the loss of the power supply.	Yes/no
	RTU can communicate simultaneously on all communication ports	Yes/no
RTU Design		
1	RTU brand and order number	
2	Double power supply Bus	Yes/no

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU
EDMS 01-300-1
24-10-2023

3	Redundant CPU	Yes/no
4	No. of port per CPU	
5	No. of digital input points	
6	No. of digital output points	
7	Switches brand and order number	
8	No. of switches	
9	No. of ports per switch	
10	Compliance with the required communication protocols	
11	NTP server brand and order number	
Signal list		
1	Compliance with signal list sheet	Yes/no
Communication		
1	Industrial routers brand and order Number	
2	The 3G/4G Router shall support LTE with bands (B1, B2, B8, B20, and B28).	Yes/no
3	No. of SIM cards per router	
4	RF modem brand and order number	
5	RF modem transmitted output power	
6	RF modem compliance with all aspects	
Cyber Security		
1	The RTU shall follow the IEC 62351 standards and at least: ☐ IEC 62351-5 : 2013 ☐ IEC 62351-3	Yes/no
2	The RTU support secure access based on role based access control.	Yes/no
3	The RTU support remote firmware updates	
4	The RTU support HTTPS or SFTP or SSH protocols for Local and remote access connection.	
	The RTU complies with industry and cybersecurity standards, implementing robust security measures such as authentication, encryption, and access controls to protect against unauthorized access and cyber threats, ensuring the integrity and reliability of the power system.	

EEHC DISTRIBUTION MATERIALS SPECIFICATION
MV Primary Switchgear RTU

EDMS 01-300-1
24-10-2023

RTU Cabinet		
1	Cabinet Dimensions	L,W,H
2	Cabinet IP	
3	Separate local/Remote switch	
4	Compliance with the design aspects	
5		
RTU Power Supply		
1	Main power supply voltage	V
2	Backup power supply voltage	V
RTU Batteries		
1	Battery Type	
2	Rated Amp. Hours Capacity	Ah
3	Manufacturer's name	
4	No. of cells forming the battery bank	
5	Nominal voltage 24 V/set	V

We guarantee the data given above for the equipment offered.

Signature:

Date: